

Destroying / Altering Tables and Views

Introduction

- In Database Management Systems, tables and views can be modified or removed using SQL commands.
- SQL provides commands to:
 - Alter structure
 - Delete objects
 - Rename objects

Altering Tables

What is ALTER TABLE?

- ALTER TABLE command is used to modify an existing table structure.

Uses of ALTER TABLE

1. Add new columns
2. Delete columns
3. Modify column datatype
4. Rename columns
5. Add constraints
6. Remove constraints

Syntax of ALTER TABLE

ALTER TABLE table_name
operation;

1. Adding a New Column

Syntax

```
ALTER TABLE Student  
ADD Email VARCHAR(50);
```

Important Points

- 7. Adds new attribute to table
- 8. Existing data remains safe

2. Modifying a Column

Syntax

```
ALTER TABLE Student  
MODIFY Name VARCHAR(100);
```

Important Points

- 9. Changes datatype or size
- 10. Used when larger storage needed

3. Dropping a Column

Syntax

```
ALTER TABLE Student  
DROP COLUMN Age;
```

Important Points

- 11. Permanently removes column
- 12. Deleted data cannot be recovered easily

4. Renaming a Column

Syntax

```
ALTER TABLE Student  
RENAME COLUMN Name TO Student_Name;
```

Important Points

- 13.Changes column name only
- 14.Data remains unchanged

5. Adding Constraints

Syntax

```
ALTER TABLE Student  
ADD CONSTRAINT pk_student  
PRIMARY KEY(Student_ID);
```

Important Points

- 15.Improves data integrity
- 16.Enforces rules on data

6. Removing Constraints

Syntax

```
ALTER TABLE Student  
DROP CONSTRAINT pk_student;
```

Important Points

- 17.Removes existing rule
- 18.Should be done carefully

Destroying Tables

What is DROP TABLE?

- DROP TABLE permanently deletes a table and its data.

Syntax

DROP TABLE Student;

Important Points

19. Removes table completely
20. Deletes all rows and structure
21. Cannot be recovered easily
22. Frees storage space

Difference Between DELETE, TRUNCATE, and DROP

Command	Removes Data	Removes Structure
DELETE	Yes	No
TRUNCATE	Yes	No
DROP	Yes	Yes

DELETE Command

Syntax

DELETE FROM Student
WHERE Student_ID=101;

Important Points

23. Deletes selected rows
24. WHERE clause can be used

TRUNCATE Command

Syntax

TRUNCATE TABLE Student;

Important Points

- 25. Deletes all rows quickly
- 26. Table structure remains

Altering Views

What is ALTER VIEW?

- Used to modify an existing view.

Syntax

ALTER VIEW Student_View AS
SELECT Name, Course
FROM Student;

Important Points

- 27. Changes view definition
- 28. Does not affect base table

Destroying Views

What is DROP VIEW?

- Used to remove a view permanently.

Syntax

DROP VIEW Student_View;

Important Points

- 29. Removes only view
- 30. Base tables remain safe

Renaming Tables

Syntax

RENAME TABLE Student TO Students;

Important Points

- 31. Changes table name
- 32. Data remains unchanged

Advantages of Altering Tables and Views

- 33. Supports database modification
- 34. Easy structure updates
- 35. Improves flexibility
- 36. Helps maintain database
- 37. Supports changing requirements

Risks of Destroying Tables and Views

- 38. Data loss possible
- 39. Relationships may break
- 40. Backup is important before DROP operation

Additional Important Points

- 41. ALTER modifies structure
- 42. DROP removes object permanently

- 43.DELETE removes selected records
- 44.TRUNCATE removes all records quickly
- 45.Views are virtual tables
- 46.DROP VIEW deletes only the view
- 47.ALTER TABLE is commonly used
- 48.Constraints improve integrity
- 49.SQL manages tables and views
- 50.Backup should be taken before destructive operations

Real-Time Example

College Database

Table

- Student

View

- Student_View

Operations

- Add Email column
- Delete unused view
- Rename table

Short Definitions for Exams

ALTER TABLE

Used to modify the structure of an existing table.

DROP TABLE

Used to permanently remove a table and its data.

DROP VIEW

Used to permanently remove a view.

One-Mark Important Points

- ALTER TABLE modifies table structure.
- DROP TABLE deletes table permanently.
- DROP VIEW removes a view.
- TRUNCATE deletes all rows only.
- DELETE removes selected rows.

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